A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMIC ANALYSIS

ANSWERS

TRINITY TERM 2017

Thursday 1st June 2017, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 6 questions in this paper.

Answer **four** *questions.*
For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

1. (a) (i) We can use the Gaussian technique to obtain

$$\left(\begin{array}{cccc|c} -1 & 3 & -2 & -6 & 2 \\ 2 & 1 & 4 & 5 & 3 \\ 1 & 2 & 2 & 1 & t \end{array}\right) \sim \left(\begin{array}{cccc|c} -1 & 3 & -2 & -6 & 2 \\ 0 & 7 & 0 & -7 & 7 \\ 0 & 5 & 0 & -5 & t+2 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{cccc|c} -1 & 3 & -2 & -6 & 2 \\ 0 & 7 & 0 & -7 & 7 \\ 0 & 0 & 0 & 0 & t-3 \end{array}\right).$$

It is now evident that:

- If $t \neq 3$, the system is inconsistent and there are no solutions.
- If $t = 3$, there are infinite solutions. From the second row, we have $x_2 = 1 + x_4$. From the first row, we get $x_1 = 3x_2 - 2x_3 - 6x_4 - 2 = 3(1 + x_4) - 2x_3 - 6x_4 - 2 = 1 - 2x_3 - 3x_4$. Therefore, solutions can be expressed as all the vectors of the form
$$\begin{pmatrix} 1 - 2x_3 - 3x_4 \\ 1 + x_4 \\ x_3 \\ x_4 \end{pmatrix}.$$

(ii) From the row echelon form analysed in the previous part:

$$\begin{pmatrix} -1 & 3 & 1 & -6 & 2 \\ 0 & 7 & 0 & -7 & 7 \\ 0 & 0 & 0 & 0 & t-3 \end{pmatrix},$$

we can divide the analysis into two cases, $t = 3$ and $t \neq 3$.
When $t = 3$:

- the rank is 2;
- the vectors cannot span $\mathbf{R}^3$ because we need three independent vectors for this;
- the collection cannot form a basis, because there are more than 3 vectors;
- no subset of vectors can form a basis because we need three independent vectors.

When $t \neq 3$: In this case,

- the rank is 3;
- the vectors span $\mathbf{R}^3$;
- the collection cannot form a basis, because there are more than 3 vectors;
- any subset of three independent vectors will form a basis
(for example, v_5 and not both v_1 and v_3).

(iii) From the previous part, we know that the rank of this set of vectors is 2. For the statement to be true, we need to reformulate it as: *Any two INDEPENDENT vectors in the collection can be extended to form a basis of* $\mathbf{R}^3$*, simply adding any vector in* $\mathbf{R}^3$ *NOT SPANNED BY THEM (OR BY THE SET).* The statement, thus, fails to be true in two types of scenarios:

- First, not every pair of vectors in the collection is independent. This is the case for vectors $\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 4 \\ 2 \end{pmatrix}$, which are linearly dependent. In this case, obviously, we cannot find any third vector forming a basis with them.
- Given a pair of independent vectors, say $\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$, the subset cannot be extended with linear combinations of these vectors, i.e. with vectors of the form $\alpha_1 \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$.

(b) F is continuously differentiable, and its partial derivative with respect to z is $F_z(x, y, z) = 6xz^2 - 3yz^2$.

(i) For the point $(x_1, y_1) = (1, 0)$, the equation becomes $1 + 2z^3 - 1 = 0$, which is solved only by $z = 0$.
$F_z(1, 0, 0) = 0$ and we cannot apply the implicit function theorem in this case.

(ii) For the point $(x_2, y_2) = (-1, -1)$, the equation becomes $-1+3-2z^3+z^3-1 = 0$, or equivalently, $z^3 = 1$, which is solved by $z = 1$.
$F_z(-1, -1, 1) = -3 \neq 0$, and we can apply the implicit function theorem.
Partial derivatives are

$$z_x(-1, -1) = -\frac{F_x(-1, -1, 1)}{F_z(-1, -1, 1)} = -\frac{5}{-3} = 5/3$$

and

$$z_y(-1, -1) = -\frac{F_y(-1, -1, 1)}{F_z(-1, -1, 1)} = -\frac{-7}{-3} = -7/3$$

2. (a) (i) Concavity can replace the second-order condition, so first-order conditions are also sufficient for an optimum.

(ii) Write $m(x) = \min\{f(x),\, g(x)\}$. Then $f(x) \geq m(x)$ and $g(x) \geq m(x)$, so:

$$\begin{aligned} f(\lambda a + (1-\lambda)b) &\geq \lambda f(a) + (1-\lambda)f(b) \\ &\geq \lambda m(a) + (1-\lambda)m(b) \\ \text{and similarly}\quad g(\lambda a + (1-\lambda)b) &\geq \lambda g(a) + (1-\lambda)g(b) \\ &\geq \lambda m(a) + (1-\lambda)m(b) \\ \text{therefore}\quad m(\lambda a + (1-\lambda)b) &\geq \lambda m(a) + (1-\lambda)m(b) \end{aligned}$$

[20%]

(b) (i) Either use the result that the sum of concave functions is concave (but should note that $\ln(x)$ is a concave function of (x, y)), or examine Hessian:

$$\begin{pmatrix} -x^{-2} & 0 \\ 0 & -3y^{-2} \end{pmatrix}$$

$\det > 0, \text{trace} < 0$, so H is negative definite on $\mathbf{R}^2_{++}$, and f is concave.

(ii) Hessian:

$$\begin{pmatrix} -12x^2 & 8 \\ 8 & -12y^2 \end{pmatrix}$$

$\det = 144x^2y^2 - 64 < 0$ when $x = y = 0$; therefore H is not negative semi-definite and hence g is not concave.

(iii)

$$h_x = y^2/(x+y)^2, \qquad h_y = x^2/(x+y)^2;$$

$$h_{xx} = -2y^2/(x+y)^3, \qquad h_{yy} = -2x^2/(x+y)^3, \qquad h_{xy} = 2xy/(x+y)^3$$

Hessian:

$$(x+y)^{-3}\begin{pmatrix} -2y^2 & 2xy \\ 2xy & -2x^2 \end{pmatrix}$$

$\det = 4x^2y^2 - 4x^2y^2 = 0, \text{trace} = -2x^2 - 2y^2 < 0$, so one eigenvalue is negative, the other is zero; hence h is concave (but not strictly so).

[30%]

(c) (i) Marginal utility tends to infinity as x or y tend to 0.

(ii) Lagrangian $\mathcal{L} = \ln x + 3\ln y + \lambda(24 - x - 2y) + \mu(24 - 2x - y)$.
K-T FOCs:

$$x^{-1} = \lambda + 2\mu \tag{1}$$

$$3y^{-1} = 2\lambda + \mu \tag{2}$$

$$\lambda \geq 0, \quad x + 2y \leq 24, \quad \lambda(24 - x - 2y) = 0 \tag{3}$$

$$\mu \geq 0, \quad 2x + y \leq 24, \quad \mu(24 - 2x - y) = 0 \tag{4}$$

(iii) • Neither constraint binds
Eqns. (3) and (4) imply $\lambda = \mu = 0$, which is inconsistent with (1) and (2).

• Both constraints bind
In this case $2x + y = 24$ and $x + 2y = 24$, so $x = y = 8$.
Substituting in (1) and (2)

$$\lambda + 2\mu = 1/24, \qquad 2\lambda + \mu = 3/24$$

This leads to $\lambda = 5/24$ but $\mu = -1/24$ which violates the non-negativity of μ.

• Only the time constraint binds
From (3), $\lambda = 0$ so (1) and (2) become

$$x^{-1} = 2\mu, \qquad 3y^{-1} = \mu$$

and hence $y = 6x$; so from (4), $24 = 2x + y = 8x$ and then $x = 3$, $y = 18$.
This violates the income constraint as $x + 2y = 3 + 36 = 39 > 24$.

• Only the income constraint binds
From (4), $\mu = 0$ so (1) and (2) become

$$x^{-1} = \lambda, \qquad 3y^{-1} = 2\lambda$$

and hence $2y = 3x$; so from (3), $24 = x + 2y = 4x$ and then $x = 6$, $y = 9$.
This satisfies the time constraint as $2x + y = 12 + 9 = 21 < 24$.
Clearly $\lambda \geq 0$ (in fact strictly so), and all the K-T conditions are satisfied.

• • Hence the only point satisfying the conditions is $x = 6$, $y = 9$.

(iv) We are maximising a concave function subject to linear constraints, so the Kuhn-Tucker conditions are sufficient, as well as necessary.
Hence the solution is optimal.

(v) A multiplier is a measure of the change in the objective function when the constraint is relaxed.
The multiplier on the income constraint is $\lambda > 0$, whereas the multiplier on the time constraint is $\mu = 0$.
Hence she would rather have a little more money.

[50%]

3. (a) The utility function u^A is continuous, strictly monotonic, and strictly quasiconcave. The demands are given by:

$$x_1^A(p, m^A) = \frac{m^A}{p_1} - \alpha, \quad x_2^A(p, m^A) = \alpha\frac{p_1}{p_2} \qquad \text{if } \frac{m^A}{p_1} \geq \alpha$$
$$x_1^A(p, m^A) = 0, \quad x_2^A(p, m^A) = \frac{m^A}{p_2} \qquad \text{otherwise}$$

In particular, x_2^A is independent of m^A when $m^A \geq \alpha p_1$.

(b) Given the endowments, $m^A = p_1 w_1^A + p_2 w_2^A > \alpha p_1$ and $m^B > p_1 w_1^B + p_2 w_2^B > \beta p_1$; therefore the excess demands for each agent are

$$z_1^A(p) = \frac{p_2}{p_1} w_2^A - \alpha, \quad z_2^A(p) = \alpha\frac{p_1}{p_2} - w_2^A$$
$$z_1^B(p) = \frac{p_2}{p_1} w_2^B - \beta, \quad z_2^B(p) = \beta\frac{p_1}{p_2} - w_2^B$$

and aggregate excess demand is

$$Z_1(p) = \frac{p_2}{p_1} w_2 - (\alpha + \beta), \quad Z_2(p) = (\alpha + \beta)\frac{p_1}{p_2} - w_2$$

Z is a function of the ratio of prices so clearly HoD 0.

Walras' Law states that $p \cdot Z(p) = 0$:

$p \cdot Z(p) = p_1 Z_1(p) + p_2 Z_2(p) = p_2 w_2 - p_1(\alpha + \beta) + (\alpha + \beta)p_1 - p_2 w_2 = 0.$

Finally, the boundary conditions are satisfied, since $\lim_{p_1 \to 0} Z_1(p) = +\infty$, and $\lim_{p_2 \to 0} Z_2(p) = +\infty$.

Therefore a Walrasian equilibrium there exists.

(c) The Walrasian equilibrium price ratio is defined by $Z(p) = 0$.

Normalising p_1 to 1, we obtain $p_2^* = (\alpha + \beta)/w_2$.

It is decreasing in w_2 and independent of w_1 (as long as $w_1^A > \alpha$, $w_1^B > \beta$).

Of course the Walrasian equilibrium *allocation* does depend on w_1.

(d) The economy behaves as if there were a representative agent with utility function $u(x_1, x_2) = x_1 + (\alpha + \beta)\ln(x_2)$ and endowment (w_1, w_2).

WARP: If $p, p' \gg 0$ are not collinear and $p\, Z(p') \leq 0$ then $p'\, Z(p) > 0$.

WARP is satisfied for individual excess demands: if $p, p' \gg 0$ are two prices that are not collinear and $pZ(p') \leq 0$, then $px(p', p' \cdot w) \leq p \cdot w$;
so at prices p the allocation $x(p', p' \cdot w)$ is affordable but demand is $x(p, p \cdot w)$ – the demand is unique (by strict quasiconcavity) so $u(x(p, p \cdot w)) > u(x(p', p' \cdot w))$.
Therefore $x(p, p \cdot w)$ cannot be affordable at price p', else it would be chosen.
Hence $p'Z(p) > 0$.

Since this economy has a representative consumer, the aggregate excess demand satisfies WARP.

(e) Uniqueness: If Z satisfies WARP, then the set of equilibrium prices is convex (and hence generically unique).
Consider two equilibrium prices p^* and p^{**}, and for $\tilde{p} = tp^* + (1-t)p^{**}$ suppose that $Z(\tilde{p}) \neq 0$.
By Walras Law $\tilde{p} \cdot Z(\tilde{p}) = 0$, and so $tp^* \cdot Z(\tilde{p}) + (1-t)p^{**} \cdot Z(\tilde{p}) = 0$.
Hence either $p^* \cdot Z(\tilde{p}) \leq 0$ or $p^{**} \cdot Z(\tilde{p}) \leq 0$.
Suppose $p^* \cdot Z(\tilde{p}) \leq 0$;
then by WARP $\tilde{p} \cdot Z(p^*) > 0$ – but this contradicts p^* being an equilibrium price.
Therefore, $Z(\tilde{p}) = 0$ and $\tilde{p}$ is an equilibrium price.

(f) Walrasian tâtonnement: $\dot{p} = Z(p)$.
The equilibrium is stable if the Walrasian tâtonnement converges.
Z satisfies the WA and hence the WA at equilibrium and so the equilibrium is stable.
Alternatively, uniqueness and only two goods implies stability.

4. (a) (i) F sosd's G if every *risk averse* expected utility maximiser prefers F to G, or if $\int u(x)\,dF(x) \geq \int u(x)\,dG(x)$ for all increasing, concave functions u, or if G is a mean-preserving spread of F.

If the CDF F crosses the CDF G from *below*, and they cross just once, then F sosd's G.
This is a *sufficient* condition, but it is not necessary.

(ii) When the possible outcomes are $X = \{-2, -1, 0, +1, +2\}$, and the lotteries are $L_1 = (1/5,\ 1/5,\ 1/5,\ 1/5,\ 1/5)$ and $L_2 = (3/20,\ 3/10,\ 0,\ 1/2,\ 1/20)$, the above sufficient condition does not hold.

Start with L_2; compound the node that results in the outcome -1 with a lottery $(\frac{1}{6}, -1; \frac{4}{6}, 0; \frac{1}{6}, +1)$, and also compound the node that results in the outcome $+1$ with a lottery $(\frac{3}{10}, -1; \frac{4}{10}, 0; \frac{3}{10}, +1)$;
this results in a lottery whose reduced version is L_1.

So, starting with L_2, to each of two possible outcome x we have added a zero-mean noise term $\tilde{z}$ distributed according to H_x,
and L_1 is the reduced lottery of this compound lottery.
So L_1 is a mean-preserving spread of L_2,
and a risk-averse expected utility maximizer prefers L_2 to L_1.

(b) Endowment is $(125, 80)$; asset pays $(15, 105)$.

(i) The agent is prepared to pay p_b for asset, where p_b satisfies:

$$\tfrac{1}{2}\ln 125 + \tfrac{1}{2}\ln 80 = \tfrac{1}{2}\ln(125 - p_b + 15) + \tfrac{1}{2}\ln(80 - p_b + 105) \quad \text{so } p_b = 60.$$

(ii) The agent already owns the asset, and is willing to sell it for p_s, where p_s satisfies:

$$\tfrac{1}{2}\ln(125 + 15) + \tfrac{1}{2}\ln(80 + 105) = \tfrac{1}{2}\ln(125 + p_s) + \tfrac{1}{2}\ln(80 + p_s) \quad \text{so } p_s = 60.$$

The buying price and the selling price are the same,
and in fact equal to the expected value of the asset $(\frac{1}{2} \times 15 + \frac{1}{2} \times 105 = 60)$.

Buying: If he were *given* the asset, then his state contingent wealth would change from $(125, 80)$ to $(140, 185)$; we need to reduce this by p_b in both states of the world to move him back onto his original indifference curve – but that point is $(80, 125)$. Essentially, he has just relabelled the 2 states.

Selling: If the asset were just *taken away*, then his state contingent wealth would change from $(140, 185)$ to $(125, 80)$; we need to increase this by p_s in both states of the world to move him back onto his original indifference curve – and that point is $(185, 140)$. Again, this is like simply relabelling the 2 states.

Consequently, $p_b = p_s$, and since he ends up neither taking on more risk nor shedding it, the buy & sell price are equal to the expected value of the asset.

Note that this conclusion does not depend on the particular functional form of the utility function. For example when buying, p_b satisfies
$\frac{1}{2}u(125) + \frac{1}{2}u(80) = \frac{1}{2}u(125 - p_b + 15) + \frac{1}{2}u(80 - p_b + 105)$ and clearly $p_b = 60$.

6. (a) The optimal portfolio problem is:

$$\max \ \ln(x_0) + \ln(x_1) + \ln(x_2)$$
$$\text{s.t. } x_0 \leq w_0 - q_\alpha z_\alpha, \quad x_1 \leq w_1 + z_\alpha, \quad x_2 \leq w_2 + 3z_\alpha$$

By monotonicity the constraints will bind and we can rewrite the problem as:

$$\max \ \ln(w_0 - q_\alpha z_\alpha) + \ln(w_1 + z_\alpha) + \ln(w_2 + 3z_\alpha)$$

FOC for A: $-\frac{q_\alpha}{1-q_\alpha z_\alpha^A} + \frac{1}{1+z_\alpha^A} + \frac{3}{3+3z_\alpha^A} = 0$ which simplifies to $z_\alpha^A = \frac{2-q_\alpha}{3q_\alpha}$.
FOC for B: $\frac{1}{z_\alpha^B} + \frac{1}{1+z_\alpha^B} + \frac{3}{1+3z_\alpha^B} = 0$ which is independent of q_α.

To compute the financial equilibrium:
From B's FOC we find z_α^B.
Market clearing implies that $z_\alpha^A = -z_\alpha^B$ and substituting in A's FOC gives q_α^*.
Replacing z_α^A, z_α^B in the budget constraints gives the equilibrium allocation.

(b) The payoff vector of β is $d_\beta = (0, 3-c)$, which is linearly independent of $d_\alpha = (1, 3)$, and so the two assets span the whole space and the markets are complete.

(c) When $c = 2$, the payoff vector for β is $d_\beta = (0, 1)$.
Markets are complete so we can solve for the corresponding endowment economy.
Let p_1, p_2 denote the prices of contingent consumptions in states 1 and 2.
Agents have Cobb-Douglas demands, hence $x_i^J = \frac{m^J}{3p_i}$:

$$x_0^A = \frac{1+p_1+3p_2}{3}, \quad x_1^A = \frac{1+p_1+3p_2}{3p_1}, \quad x_2^A = \frac{1+p_1+3p_2}{3p_2}$$

$$x_0^B = \frac{p_1+p_2}{3}, \quad x_1^B = \frac{p_1+p_2}{3p_1}, \quad x_2^B = \frac{p_1+p_2}{3p_2}$$

The market clearing conditions are: $x_0^A + x_0^B = 1, x_1^A + x_1^B = 2, x_2^A + x_2^B = 4$
implying the state prices are $p_1 = 1/2, p_2 = 1/4$
and the consumption allocations are $x^A = (3/4, 3/2, 3)$, $x^B = (1/4, 1/2, 1)$.

Given the state prices, asset prices are $q_\alpha = p_1 + 3p_2 = 5/4$ and $q_\beta = p_2 = 1/4$.
Portfolio choices are $(z_\alpha^A, z_\beta^A) = (1/2, -3/2)$, $(z_\alpha^B, z_\beta^B) = (-1/2, 3/2)$.

(d) The price vector (q_α, q_β) is arbitrage free if there is no portfolio $z = (z_\alpha, z_\beta)$ such that $q_\alpha z_\alpha + q_\beta z_\beta \leq 0$ but $z_\alpha \geq 0$ and $3z_\alpha + z_\beta \geq 0$.

In this case $(q_\alpha, q_\beta) = (1, 1)$ is not arbitrage free, since $z = (1, -1)$ has cost 0 and positive returns in any state.

(e) The asset γ is a linear combination of α and β: $d_\gamma = d_\alpha - 2d_\beta$.
Therefore the only price that does not lead to arbitrage is $q_\gamma = q_\alpha - 2q_\beta = 4 - 2 = 2$.

5.

(a) $\pi = 4q_1 + 5q_2 - (q_1+q_2)^2/10$

$\frac{\partial \pi}{\partial q_1} = 4 - (q_1+q_2)/5 \geq 0 \Leftrightarrow q_1+q_2 \leq 20.$ ✓

$\frac{\partial \pi}{\partial q_2} = 5 - (q_1+q_2)/5 \geq 0 \Leftrightarrow q_1+q_2 \leq 25$ ✓

$\therefore \underline{\underline{q_1 = q_2 = 10}}, \quad \pi = 90 - 40 = \underline{\underline{50}}$

(b) Either $p = 4, \quad \pi = 4(q_1+q_2) - (q_1+q_2)^2/10$

$\frac{\partial \pi}{\partial q_i} = 4 - (q_1+q_2)/5 \geq 0 \Leftrightarrow q_1+q_2 \leq 20$ ✓

$\therefore q_1 = q_2 = 10, \quad \pi = 80 - 40 = \underline{\underline{40}}$

or $p = 5, \quad \pi = 5q_2 - q_2^2/10$

$\frac{\partial \pi}{\partial q_2} = 5 - q_2/5 \geq 0 \Leftrightarrow q_2 \leq 25$ ✓

$\therefore q_1 = 0, q_2 = 10, \quad \pi = 50 - 10 = \underline{\underline{40}}.$

$\therefore$ indiff. between $p=4$ & selling to both, & $p=5$ & selling to high.

(c) Can always replicate (b) with $F=0$, $p=p^*$.

Can always replicate (b) with $F=0, p=p^*$ as long as this results in

- if $p \leq 4$, both buy 10 as long as this results in non-neg. consumer surplus;
  so $F=0, p=4$ is o.k., or even $F=40, p=0$.
- if $4 < p \leq 5$, high buys 10, low buys 0, as long as c.s. ≥ 0;
  so $F=0, p=5$ is o.k., or even $F=50, p=0$.
- if $5 < p$, neither buys.

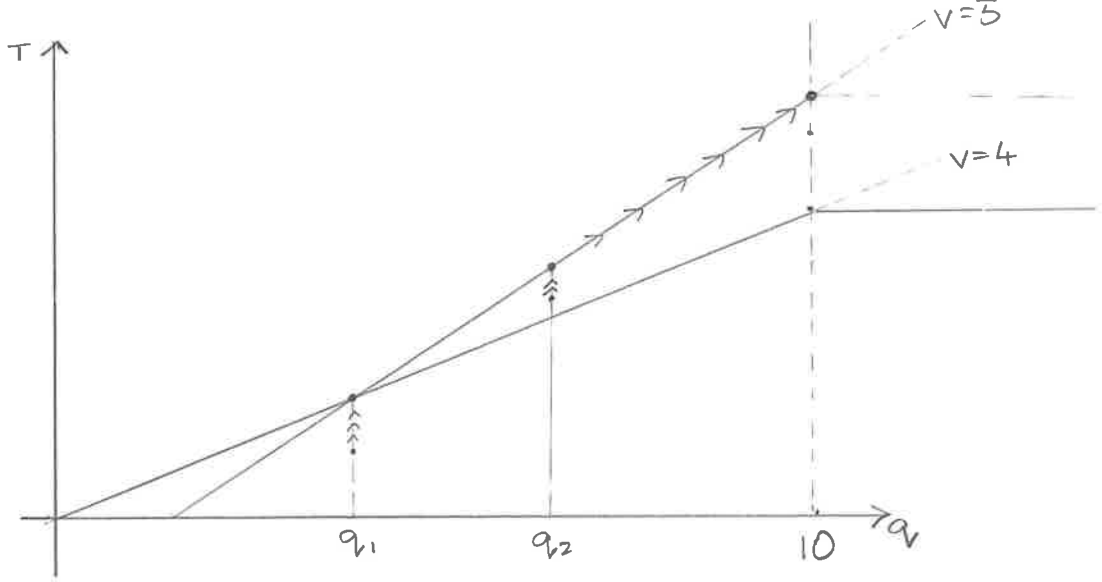
She is indifferent between the first two, & these replicate (b).

[Note: We have assumed $F \geq 0$. If $F < 0$, the consumer can take the F & buy 0, which is loss-making.]

(d) Can always replicate (c) with $\langle q_i(p^*), F + p^* q_i(p^*) \rangle$

5 (ctd)

(d)



L contract is <u>on</u> (or below) $v=4$ thro' origin,

i.e. $\langle q_1, 4q_1 \rangle$

H contract is <u>on</u> (or below) $v=5$ thro' L contract,

i.e. $\langle q_2, 4q_1 + 5(q_2 - q_1) \rangle$. $= \langle q_2, 5q_2 - q_1 \rangle$

$$\pi = 4q_1 + (5q_2 - q_1) - (q_1 + q_2)^2/10$$

$$\frac{\partial \pi}{\partial q_2} = 5 - (q_1 + q_2)/5 \geq 0 \iff q_1 + q_2 \leq 25.$$

$\therefore q_2 = 10$ (regardless of q_1, given $0 \leq q_1 \leq 10$),

& H contract is $\langle 10, 50 - q_1 \rangle$

Now, $\pi = 4q_1 + (50 - q_1) - (q_1 + 10)^2/10$

$$\frac{\partial \pi}{\partial q_1} = 3 - (q_1 + 10)/5$$

$= 0$ when $q_1 + 10 = 15$, $q_1 = 5$. (soc o.k. for max.)

$\therefore$ L contract is $\langle 5, 20 \rangle$

& now H contract is $\langle 10, 45 \rangle$.

$$\pi = 65 - 22.5 = \underline{\underline{42.5}}$$